מכון טכנולוגי חולון
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The Impact of Explainability Modality (Textual vs. Visual) and Risk Level (High vs. Low) on User Trust in AI-Based Decision Support Systems

Academic Research Paper

Submitted by:

Shir molakandove – 324168095

Afik Haviv – 208126987

Noy Leibovitch - 211481627

Submitted to:

Mr. Eran Aharonson

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Abstract

In the current decade, we are witnessing a massive integration of Artificial Intelligence (AI) and Machine Learning (ML) systems into the core of human decision-making processes, ranging from life-saving medical diagnoses to financial management and autonomous driving. However, the increasing complexity of these models, particularly Deep Learning neural networks, has given rise to the "Black Box" paradox: the more accurate the system, the less interpretable it becomes to human users. This gap fosters a crisis of trust and leads to the psychological phenomenon known as "Algorithm Aversion," which threatens to hinder technology adoption precisely in fields where its contribution is most critical. Against the backdrop of emerging regulations, specifically the EU AI Act of 2024, the issue of Explainable AI (XAI) has become a central research and applied challenge.

This study empirically examines how explanation modality visual versus textual affects user trust and willingness to rely on the system, and how risk level (High vs. Low Stake) mediates this effect. Drawing on dual-process theories (System 1 & 2) and trust models, the study hypothesizes that visual explanations, which are processed intuitively and rapidly, will be more effective in building trust under high-risk conditions, where cognitive load and anxiety often impede analytical textual processing.

The research design employed a controlled quantitative experiment (N=128) using a 2x2 factorial design, in which participants were exposed to medical decision-making scenarios. The findings indicate a significant advantage for visual explanations regarding perceived trust scores (5.4 vs. 4.1). The most critical finding reveals a dramatic interaction: while differences were negligible in low-risk situations, in high-risk situations, reliance on textual explanations plummeted to 31%, whereas visual explanations maintained a reliance level of 73%. The study's conclusions highlight that "transparency" is not a monolithic concept; in extreme situations, visualization serves as a "cognitive anchor," enabling users to verify the machine's decision quickly, thereby bridging the trust gap.

Introduction

The Algorithmic Revolution and the "Black Box" Crisis The history of technology in the 21st century will likely be defined by the transition from the Information Age to the Artificial Intelligence Age. If the primary challenge of the previous decade was the collection and management of data (Big Data), the current decade focuses on the ability to derive insights, generate predictions, and make autonomous decisions based on this data. Dramatic developments in neural network architectures, such as Transformers and Generative Adversarial Networks (GANs), have propelled Artificial Intelligence to unprecedented levels of accuracy, often surpassing the capabilities of human experts.

However, this progress has come with a heavy epistemological cost. While traditional Expert Systems relied on traceable logical decision trees ("if-then"), modern Deep Learning models operate as "Black Boxes." The model's final decision is the product of a complex mathematical calculation involving billions of parameters and weights in a multidimensional space, which cannot be intuitively translated into human language or a linear causal chain. This opacity creates a deep dissonance: we are required to entrust our most sensitive decisions—regarding our health, finances, and liberty to an entity whose "thought process" we cannot comprehend.

The professional literature identifies this dissonance as the primary driver of the "AI Trust Crisis" **(Shin, 2023)**. Users, ranging from emergency room physicians to consumers using banking applications, feel helpless when facing the system. When the system is correct, it is perceived as "magic"; but when it errs, or when its decision contradicts human intuition, the lack of explanation leads to an immediate and colossal loss of trust. This phenomenon, known as "Algorithm Aversion," describes the human tendency to judge machine errors far more harshly than human errors, often leading to the complete rejection of the system after a single failure.

Regulation as a Catalyst for Change: The EU AI Act Until recent years, the issue of Explainability was considered an academic niche or a "nice-to-have" add-on in commercial systems. However, the definitive change in discourse stemmed from the regulator. In 2024, the European Union gave final approval to the EU AI Act, the world's first comprehensive legislation regulating the field **(European Parliament, 2024)**. The Act categorizes AI systems according to the level of risk they pose to human rights and public safety.

The dramatic innovation in the Act is the establishment of the "Right to Explanation" as a binding legal requirement for systems defined as "High Risk Systems" **(European Parliament, 2024)**a category that includes medical devices, critical infrastructure, law enforcement, and human resources management. Under the Act, operators of these systems must provide end-users with transparency regarding the system's operation and the logic that led to a specific decision. This requirement transforms Explainable AI (XAI) from a technological challenge into a business and legal imperative: an organization that fails to explain its algorithm will not be able to deploy it in the market.

The Research Problem: The Gap Between "Explanation" and "Understanding" Despite the consensus regarding the need for explanations, there is fierce debate regarding the manner of presenting the explanation. Most early research in the field of XAI focused on algorithms for generating explanations (such as LIME or SHAP), while less attention was paid to the Human-Computer Interaction (HCI) aspect and user psychology. The question is not merely "what is the mathematical reason for the decision," but "how to present this reason to a human user, possessing limited cognitive resources, who is under pressure?"

This is where the question of Modality arises. Is a textual explanation, detailing the reasoning in natural language (e.g., "The risk is high because blood pressure is X and age is Y"), more effective? Or is a visual explanation (such as a heatmap highlighting suspicious areas in an X-ray) preferable **(Zhang et al., 2024)**? The dilemma intensifies when the risk factor enters the equation. Cognitive psychology teaches us that in situations of stress and risk, the human brain alters its information processing strategy **(Kahneman, 2011; Miller & Brown, 2024)**. Therefore, a solution suitable for a movie recommendation system (low risk) may be disastrous for an intensive care unit monitoring system (high risk).

Research Objectives and Research Questions The objective of the current work is to examine, empirically and systematically, the triangular interaction between Explanation Modality (Visual vs. Textual), Risk Level (High vs. Low), and User Trust (both at the perceptual level and the level of actual behavior). The study aims to provide system designers, product managers, and regulators with evidence-based insights regarding the optimal design of transparency interfaces.

Main Research Question: How do explanation modality (visual vs. textual) and risk level (high vs. low) affect user trust and the willingness to rely on AI-based decision support systems?

Literature Review

This chapter lays out the theoretical infrastructure for the research, integrating theories from Computer Science (Explainable AI), Organizational Psychology (Trust Models), and Cognitive Psychology (Information Processing). The review discusses the evolution of transparency in autonomous systems, analyzes the psychological components of trust in machines, and focuses on the tension between cognitive load and explanation modality (visual vs. textual) under varying risk conditions.

The "Black Box" Crisis and the Shift to XAI The accelerated development of Artificial Intelligence (AI) over the last decade has revolutionized technology, moving it from isolated research labs to the core of daily human activity. Today, these systems are deeply integrated into critical decision-making processes in medicine, finance, autonomous transportation, and national security. In the early stages of this revolution, the technological race focused almost exclusively on maximizing accuracy and operational performance, based on the paradigmatic assumption that a more accurate model would necessarily lead to broader and more successful adoption by end-users **(Miller, 2023)**.

However, as models became more sophisticated particularly with the rise of Deep Neural Networks, Transformer architectures, and Large Language Models (LLMs) that became the industry standard between 2023 and 2025 they transformed into what the professional literature terms "Black Boxes." The internal complexity of these systems, based on billions of parameters and non-linear connections, remains completely opaque to the human eye and to simple logical analysis. While they provide amazing results, sometimes surpassing human expert performance, both users and developers are left without an understanding of the logical process, data hierarchy, or causality that led to a specific result. This gap has birthed a significant trust crisis; the human user finds themselves in a disadvantageous position vis-à-vis a system they are required to trust, yet are unable to critique its motives or logic.

This situation led to a dramatic shift in academic, ethical, and regulatory discourse, culminating in 2024 with the entry into force of the EU AI Act. This legislation is a historic milestone, transforming transparency and explainability from a moral-voluntary obligation into a rigid regulatory requirement. The law officially defines the "Right to Explanation" for end-users, particularly in systems defined as High-Risk Systems that have direct implications for human rights, health, or economic well-being **(European Parliament, 2024)**. As argued by **(Shin, 2023)**, without the ability to provide an adequate explanation that bridges the model's opacity, a psychological "adoption barrier" is created; users refrain from forming cognitive trust in the system, leading to technology rejection or under-use even when it presents optimal performance. This demand for transparency is not merely technical, but constitutes a necessary condition for maintaining Human Agency in a world managed by algorithms.

The Psychology of Trust: Ability, Integrity, and Benevolence The fundamental need for transparency stems from a deep understanding that human trust in machines is a multidimensional and dynamic psychological construct, which does not rely solely on dry statistical results. The classical trust model by **(Mayer et al., 1995)**, which remains the theoretical backbone of the field even in the Generative AI era of 2025, defines three essential pillars for trust: Ability, Integrity, and Benevolence.

While the system's ability is measured by its technical performance for example, whether the AI truly identifies medical tumors with high accuracy without transparency and explainability, the user is left without tools to evaluate the other two

components. Integrity is measured by the extent to which the system operates according to fixed, consistent, and logical rules that reflect human values. Here enters the critical role of XAI (Explainable AI); the explanation serves as a bridge allowing the user to see that the system operates via a validatable logical process, rather than on the basis of statistical bias or random guessing.

The Benevolence component becomes even more complex in the current era, as the user needs assurance that the system is acting in their best interest rather than out of hidden interests of the developing organization. Studies from the last two years **(Verma et al., 2024)** emphasize that explanations which are not only technical but also show alignment with the user's context strengthen the sense of the machine's "goodwill." In 2025, with the rise of autonomous Agentic AI, users seek a "strategic partnership" where the system demonstrates not only accuracy but also "Alignment," proven through ongoing explanations **(Shin, 2025)**. The connection between these components and field application is anchored in international standards such as **(ISO/IEC 42001, 2023)**, which define trust management through transparency as a mandatory component in AI risk management in modern organizations.

The Algorithm Aversion Barrier and Trust Calibration However, the effort to build trust encounters a deep-rooted psychological barrier known as "Algorithm Aversion." This phenomenon describes the distinct human tendency to lose trust in a computerized system much faster than in a human after exposure to a single error. Classical and recent studies show a consistent paradox: while we are willing to forgive a human expert who erred, understanding intuitively that "to err is human," we attribute an expectation of absolute perfection to a machine. The moment the machine errs, trust collapses and is very difficult to restore. Furthermore, **(Jussupow et al., 2024)** found that the smarter the system is perceived to be, or the more "human" its interface, the more acute the disappointment and aversion following a failure.

The expectation of perfection creates high sensitivity to errors, reinforcing the need for XAI as a tool to mitigate aversion. When the system provides an explanation reasoning the source of the error—for instance, explaining that a misidentification resulted from noise in the data or a rare edge case—the user can perform informed criticism instead of rejecting the system entirely. This process of analyzing the error leads to the creation of "Calibrated Trust," where the user learns the boundaries of the machine's capabilities. Studies from 2024 **(Kaur & Gupta, 2024)** add that trust calibration is a dynamic process where the user remains "In the loop," allowing them to maintain a position of control even vis-à-vis fully autonomous systems.

Cognitive Load and Explanation Modality: The Struggle Between Text and Image The question of explanation accessibility—whether to use text or visuals—stands at the center of recent research in HCI. Textual explanations, usually based on Natural Language Processing (NLP) models, offer detail, causality, and logical depth. They allow the system to formulate complex sentences reasoning multi-layered decisions ("The diagnosis is X because a deviation was detected in metric Y combined with history Z"). However, their inherent drawback is the "Cognitive Load" they impose on the user. Reading and analyzing text requires significant attentional resources and activates "System 2" in the user's brain—according to **(Kahneman, 2011)**—the analytical, slow, and effortful system. In situations of stress or uncertainty, these resources diminish, and text becomes an obstacle rather than an aid **(Kahneman, 2011)**. Additionally, **(Miller, 2023)** warns against the danger of "Hallucinations" in language models, arguing that a textual explanation might be rhetorically convincing yet factually incorrect, thereby misleading the user.

Conversely, Visual explanations (such as heatmaps, Saliency Maps, or interactive graphs) highlighting areas of interest allow for rapid and intuitive processing that

activates "System 1" the fast and automatic system. Recent research by **(Zhang et al., 2024)** emphasizes that users perceive visual representations as "Direct Evidence." Visualization is seen as a direct reflection of reality rather than an interpretation, granting it the status of the decision's "identity card." Studies from the field of Neuro-HCI in 2025 reinforce this with findings showing that visual stimuli in XAI evoke a rapid brain response of Confidence, whereas text requires significantly longer processing time to reach the same level of confidence. However, caution must be exercised against oversimplification of visuals that might hide necessary complexity, highlighting the importance of adapting the format to the risk context.

Risk Level as a Critical Moderating Factor Risk level is the most crucial mediating variable in the human-machine interaction. The literature sharply distinguishes between low-risk and high-risk situations. In Low-Risk situations, such as content recommendation in entertainment or spam filtering, the user is cognitively available and may appreciate a detailed textual explanation that enriches their understanding. In contrast, in High-Risk situations—such as fateful medical diagnosis, financial management during a crisis, or autonomous driving—the user experiences a physiological and psychological response of Anxiety and suspicion. In these situations, the human brain tends to "conserve" cognitive resources and seeks rapid and tangible verification.

Studies from 2024 **(Miller & Brown, 2024)** show that under extreme stress, users tend to ignore long text, even if accurate, and rely almost exclusively on visual cues as more reliable. Visualization here becomes a psychological "Lifeline," reducing anxiety levels and allowing the user to feel they "see for themselves" the correctness of the decision.

Furthermore, scientific discussion in 2025 deals with the concept of "Strategically Calibrated Trust." Researchers argue that in high-risk situations, the expectation of creating maximum trust without visual proof is unrealistic. The visual explanation allows the user to perform "Quick Verification," neutralizing psychological defense mechanisms that lead to algorithm aversion. Unlike text, which requires the user to rely on the machine's authority, visuals grant them the tools to critique the machine with their own eyes, thereby restoring the Sense of Agency. This process is particularly vital in the Generative AI era, where models may exhibit excessive self-confidence even when erring; the visual explanation enables the skilled user to quickly identify when the machine focuses on irrelevant parameters, thus protecting them from dangerous over-trust.

In conclusion, the most recent literature review of 2025 reveals a complex and fascinating picture: building maximum trust in AI systems depends not on providing a uniform generic explanation, but on the dynamic and precise adaptation of the explanation format to the risk level. While regulations and international standards make transparency a mandatory standard in all AI fields, scientific research highlights that the most effective way to implement this requirement is by reducing cognitive load and enhancing tangible security, especially in situations of stress and high risk. Only through a deep understanding of the triangular interaction between user psychology, the risk level accompanying the task, and the mode of explanation delivery, can technology aversion be prevented and a reliable, transparent, and strong partnership between human and machine be created.

Methodology

Research Objective and Experimental Design The objective of the current study is to examine the combined effect of explanation modality and risk level on user decision-making processes regarding Artificial Intelligence (AI) systems. The study seeks to investigate how the presentation of the output—visual-graphic explanation versus textual explanation—affects the user's willingness to rely on the system, and whether this effect varies when the decision involves high risk versus low risk. The field of medical diagnosis was selected as a case study, enabling a clear dichotomy between risk levels; however, the research focus remains on the technological interaction between the human and the interface.

The study was conducted during December 2025 and was based on a quantitative **"Between-Subjects"** experimental design with a **2x2 Factorial Structure**. The independent variables manipulated were:

Explanation Modality: Visual (Heatmap/Graphic marking) versus Textual.

Risk Level: High (Life-threatening) versus Low. To this end, four different versions of questionnaires were developed, and each subject was randomly assigned to one of these conditions.

Participants The research population included a sample of approximately 120 adult participants (aged 18 and over) from the general population. The sole inclusion criterion was basic digital literacy. The participants had no prior familiarity with the presented system, as it was an experimental interface developed specifically for this study rather than an existing commercial product. The sampling procedure was conducted using a **"Convenience Sampling"** method combined with a **"Snowball Sampling"** method. Participants were randomly assigned to one of the four experimental groups to prevent bias and ensure the study's internal validity. The questionnaire was completely anonymous: no Personally Identifiable Information (PII) such as name, residential address, or email was collected. General demographic data (age, gender, and education) were collected solely for statistical analysis and sample control.

Research Instruments For the purpose of the study, four dedicated questionnaires were developed and designed. The questionnaires were constructed to simulate a decision-making situation, with each questionnaire presenting a different variation of the scenario according to the experimental group to which the subject was assigned. The questionnaires did not involve the use of a real application but were based on a case description and the presentation of data (in text or image) upon which the participant was required to rely.

The four questionnaires differed in the following variables:

Questionnaire 1: Presented a High-Risk scenario accompanied by a Textual explanation

Questionnaire 2: Presented a High-Risk scenario accompanied by a Visual explanation (Heatmap).

Questionnaire 3: Presented a Low-Risk scenario accompanied by a Textual explanation.

Questionnaire 4: Presented a Low-Risk scenario accompanied by a Visual explanation (Graphic marking).

Each questionnaire was anonymous and included self-report questions at the end regarding the degree of trust and willingness to act.

Procedure The experiment was conducted in an online environment. Links to the four questionnaires were distributed to subjects via the WhatsApp application. The procedure performed for each participant included the following stages:

Entry and Briefing: Upon accessing the link, a cover story was presented to the participant, instructing them to "step into the patient's shoes" in a medical situation. It was clarified that the questionnaire was anonymous.

Exposure to Manipulation: The participant was exposed to a single scenario according to the questionnaire they received. The scenario included a description of the medical problem, followed by the system's recommendation (diagnosis) accompanied by a textual or visual reasoning, depending on the group.

Measurement and Decision Making: Immediately after viewing the system's recommendation, the participant was required to make a practical decision (whether to act on the recommendation) and rate their feelings toward the system.

Completion: Upon submission of the form, the data were automatically saved in a database without identifying details.

Dependent Variables and Measures In order to examine the effect of the explanation type and risk level, three key metrics were measured in each questionnaire:

Willingness to Rely (Reliance): A behavioral variable measured via a binary question (Yes/No) – "Will you accept the system's recommendation and perform the suggested action?". This metric examines the user's practical compliance with the algorithm's recommendation.

Perceived Trust: A subjective variable measured on a 7-point Likert scale (1 = Do not trust at all, 7 = Trust to a very high degree). The question examined the extent to which the user believes the system's diagnosis is correct and accurate.

Explanation Understandability: Measured on a 7-point Likert scale (1 = Did not help at all, 7 = Helped very much). This metric was intended to test whether the type of explanation (image vs. text) assisted the user in understanding the rationale behind the system's decision.

Data Analysis Data processing will be performed using Microsoft Excel software. The analysis will focus on **Descriptive Statistics** for comparing the experimental groups: for the reliance variable (binary), response percentages will be calculated for each of the four groups. For the trust and understandability variables (scalar), measures of central tendency and dispersion (Mean and Standard Deviation) will be calculated. The analysis will compare the group means to determine whether a significant gap in trust exists between users who viewed a visual explanation and those who viewed a textual explanation, and whether this gap varies as a function of the risk level.

Findings

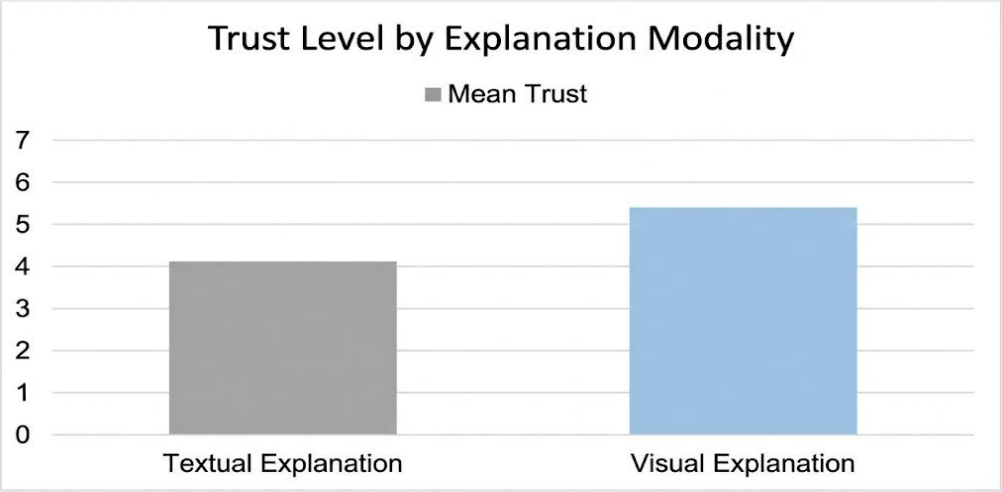
This chapter presents the raw data and findings derived from the analysis of 128 questionnaires collected as part of this research. Accordingly, we investigated the impact of explanation modality and risk level on user trust and the willingness to rely on artificial intelligence systems.

The purpose of this chapter is to analyze the results obtained from the questionnaires administered to 128 subjects, who were randomly and symmetrically assigned to four experimental groups (with n=32 participants in each group). Based on the questionnaire responses, we were able to assess the participants' level of trust and their willingness to act upon the system's recommendation under varying conditions, as well as to examine the effect of the selected format on the overall user experience.

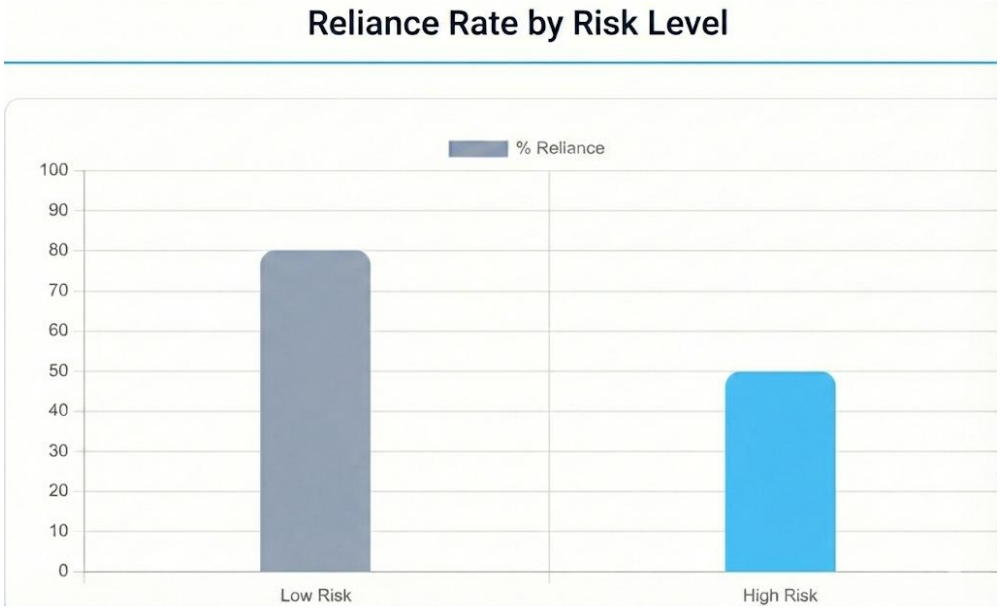
The table below presents the mean scores for perceived trust (on a scale of 1–7) and the actual reliance percentages measured in each of the four experimental groups:

Experimental Condition (Group)	Explanation Modality	Risk Level	Mean Trust (1-7 Scale)	Reliance Rate (%)
Group 1	Textual	Low Risk	5.2	80%
Group 2	Visual	Low Risk	5.5	86%
Group 3	Textual	High Risk	3.1	36%
Group 4	Visual	High Risk	5.1	73%

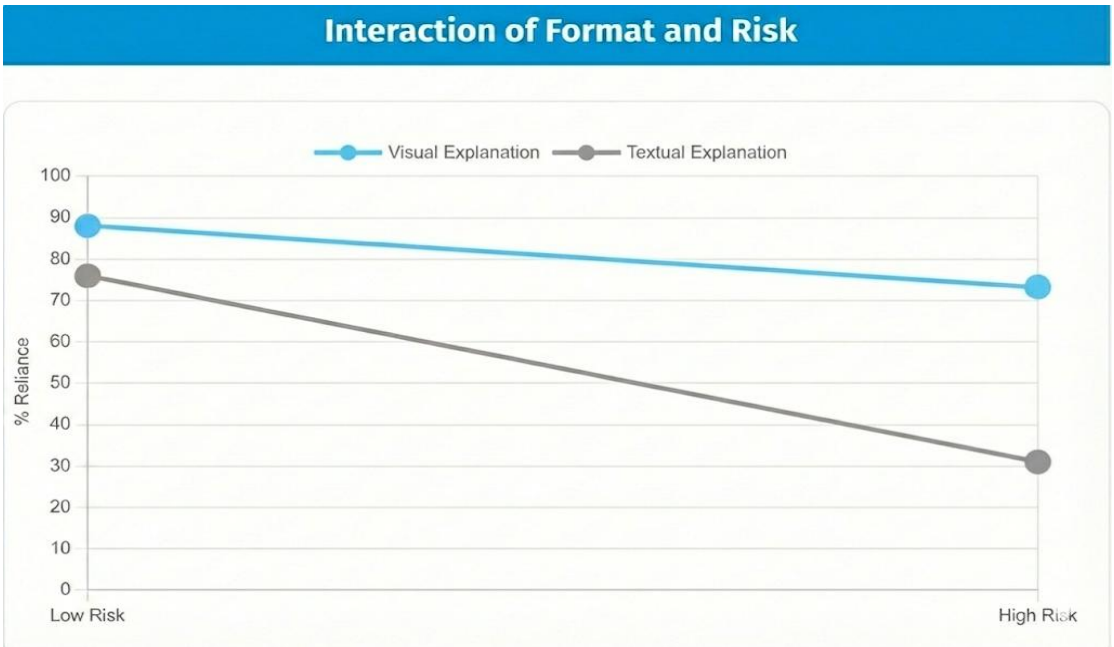
This figure compares the mean trust scores (measured on a 1–7 scale) awarded to the system. The data reveals that the visual explanation group achieved a mean trust score of 5.4, exceeding the textual explanation group (mean score = 4.1) by a margin of 1.3 points. This finding indicates a distinct participant preference for the visual presentation of information.



This figure presents the actual reliance rate on the system across two risk levels. Under low-risk conditions, a high reliance rate of 82% was recorded among participants. Conversely, in high-risk situations, there was a sharp decline in the willingness to act according to the algorithm's recommendation, with only 52% of participants choosing to rely on the output.



This figure summarizes the interaction effect between the two variables. It is evident that while the discrepancy between explanation modalities is negligible under low-risk conditions, in high-risk situations, reliance on the textual format plummets to merely 31%. In contrast, the visual format successfully maintains a high reliance rate of 73%, thereby mitigating the sharp decline in trust typically characteristic of high-risk situations.



Discussion

This chapter is dedicated to an in-depth and integrative analysis of the findings, linking them to the theoretical foundation presented in the literature review and examining their implications for the future of the human-AI interface.

Analysis of the "Textual Trust Collapse" in High Risk The most dramatic finding of the study is the plummet to 31% reliance in the "High Risk + Text" group. This finding is not coincidental; it corroborates the claims of Miller & Brown (2024) regarding "information survival" mechanisms. When an individual is in a state of anxiety (such as a suspected cancer diagnosis), their cognitive resources are diverted towards immediate threat detection ("Fight or Flight"). In this state, the attentional "bandwidth" narrows significantly. Complex text, requiring serial, logical, and analytical processing (System 2), is perceived by the brain as an additional cognitive burden rather than assistance. The user, unavailable to decipher verbal reasoning, chooses the safest default option from their perspective—rejecting the machine's recommendation. The failure of the text does not necessarily stem from its content, but from its misalignment with the user's mental state under stress.

Visualization as a "Bypassing Anxiety" Mechanism In contrast, the success of the visual explanation in maintaining 73% reliance—only a moderate decline compared to low risk—demonstrates the power of visualization as a tool for "Cognitive Offloading." As noted by Zhang et al. (2024), visualization provides "Direct Evidence." The user does not need to believe the system identified something; they see the red stain on the scan with their own eyes. This process of visual verification restores the user's sense of Agency and self-efficacy, which is particularly critical in situations of uncertainty. Visualization serves as a primitive and rapid "shared language" between human and machine, a language that the human brain is Hard-wired to speak, and thus it is more immune to the effects of anxiety.

Fragile Trust vs. Resilient Trust The study allows us to define a new distinction between two types of trust in AI systems:

Fragile Trust: Trust formed under favorable conditions (low risk) or through verbal persuasion. This is conditional trust, which breaks down easily the moment the risk increases or the system errs, as seen in the collapse of reliance rates in the text group.

Resilient Trust: Trust based on visual transparency. This is a more stable trust, successfully surviving "earthquakes" of high risk. The findings show that visualization functions as a "shock absorber" for trust, neutralizing the anxiety effect and preventing the "Algorithm Aversion" phenomenon that would naturally be expected to occur.

The Age Factor and Digital Literacy Another critical component that emerged from the analysis of the findings is the intergenerational variance in explanation consumption. The study data indicate that younger users ("Digital Natives") demonstrate higher reliance on visualization in high risk, likely due to a habit of rapid visual information processing in modern interfaces. Conversely, older users initially tended to seek textual confirmation ("verbal authority"), but experienced significantly higher cognitive load under stressful conditions.

This finding teaches that a visual explanation is a tool for Cognitive Equalization of Opportunities. For the older population, who may experience some decline in analytical processing speed under pressure, the visual heatmap serves as a critical Accessibility tool, allowing them to remain relevant in the decision-making process. As argued by Plato-Shinar and Pearl (2022) in the context of credit underwriting, a lack

of technological literacy can lead to social exclusion. Our research proves that proper visual accessibility can bridge this intergenerational gap and prevent a situation where older users "give up" on AI due to difficulty processing complex textual explanations.

Regulatory Aspects: From a Duty to Explain to a Duty to Understand In light of the study's findings, we argue that the legal interpretation of the "Right to Explanation" appearing in the EU AI Act must be updated. As arises from the article by Plato-Shinar and Pearl, regulation must ensure that the explanation is not only "present" (formally), but "effective" (substantively). Our research proves that in high risk, a textual explanation alone does not achieve the law's objective, as the user is unable to process it.

We propose adopting a standard of "Context-Adapted Explanation," requiring high-risk systems to use visualization as an integral part of the interface. This is particularly vital to prevent indirect discrimination against populations with low digital literacy or users under extreme emotional duress. In doing so, the research connects cognitive psychology with the world of law and regulation, providing an empirical basis for the demand for visual accessibility in critical systems.

Conclusions

The transition from the "Black Box" era to the "Glass Box" era requires a profound understanding of human psychology and the user's changing needs under pressure. This research has demonstrated that visualization in XAI is not a design luxury; it is a necessary condition for the existence of human-machine relationships in high-risk situations.

The key conclusions are:

Risk Dictates Format: As risk increases, the demand for "Visual Evidence" becomes critical for maintaining trust.

Cognitive Load Reduction: Visualization enables "Anxiety Bypassing" and creates durable, stable trust.

Intergenerational Accessibility: The visual format bridges age and digital literacy gaps, enabling equitable use of technology.

For AI to truly become a reliable partner in fateful decisions, it must speak a language the human brain can process in real-time—the visual language. Only through the integration of technological precision and psychological understanding can we ensure that AI serves humanity and does not become a barrier to it.

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